MULTILINGUAL VOICE-BASED PUBLIC SERVICE ASSISTANT

A MINI PROJECT REPORT

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NOVEMBER 2026

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ACKNOWLEGEMENT

We express our sincere thanks and profound gratitude to Dr.K. P. Ramasamy, Chairman, and Ms. Gayathri Anandakrishnan, Secretary, KPR Group ofInstitutions, for providing an excellent academic environment and for.their visionary leadership and unwavering support for research and innovation.

We are deeply grateful to Dr. M. Akila, Principal, KPR Institute of Engineering and Technology, for her esteemed guidance and for fostering a culture of academic excellence.

We are especially thankful to [Name of HoD], Professor and Head, Department of Artificial Intelligence and Data Science, for his/her support. We owe our deepest gratitude to our project supervisor, [Name of Supervisor], Assistant Professor, Department of Artificial intelligence and data science Intelligence ,whose expert guidance, immense patience, and constant encouragement have been the cornerstones of this endeavour.

We also extend our heartfelt thanks to our beloved family, faculty members, supporting staffs and friends for their unwavering support and motivation.

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ABSTRACT

Millions of citizens find it difficult to discover and use government services because information is spread across many portals, is often available only in a few languages, and assumes a level of digital literacy that many people do not have. Language barriers and complex navigation prevent citizens from learning about schemes, certificates, healthcare, transport, education and job services, and from filing applications or complaints correctly. This project proposes a Multilingual Voice-Based Public Service Assistant that allows citizens to ask questions by speaking in their preferred Indian language. The spoken query is converted to text by a speech recognition module, the language is detected, and an AI/NLP module understands the citizen's intent rather than relying on keywords alone. The system then identifies the relevant government service and department, retrieves verified service information, and generates simple step-by-step guidance covering eligibility, required documents, procedure, fees, timelines and nearby service centres. The guidance is translated into the citizen's own language and delivered back as natural speech through text-to-speech. By combining voice recognition, language technology, AI-based query understanding and public-service navigation on a single platform, the proposed assistant offers accessible, inclusive and always-available guidance, particularly for elderly people, low-literacy users and those uncomfortable with English or text-based portals. The design is modular and can be extended with more languages and integrated with verified real-time government APIs and databases, making it a step towards inclusive and citizen-centric digital governance.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
CSS	Cascading Style Sheets
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
HTTPS	HyperText Transfer Protocol Secure
MeitY	Ministry of Electronics and Information Technology
NIC	National Informatics Centre
NLP	Natural Language Processing
RAM	Random Access Memory
SSD	Solid State Drive
TTS	Text-to-Speech
UMANG	Unified Mobile Application for New-age Governance
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 OBJECTIVE

The main objective of this project is to make public-service information accessible to every citizen in their own language through natural voice interaction. The specific objectives are:

1. To develop a multilingual AI voice assistant that allows citizens to communicate in their preferred Indian language.
2. To understand voice-based queries related to government services, schemes, certificates, healthcare, transport, education, jobs, and complaints.
3. To provide simple and accurate step-by-step guidance about eligibility, required documents, application procedures, and service centres.
4. To reduce language and digital-literacy barriers by providing information through natural voice interaction.
5. To make public-service information easily accessible, user-friendly, and available anytime through a single AI-powered platform.

1.2 OVERVIEW OF MULTILINGUAL VOICE-BASED PUBLIC SERVICE ASSISTANCE

Governments provide a wide range of services such as welfare schemes, certificates, healthcare, transport, education and employment support. Although many of these services are now available online, a large section of citizens still struggles to reach them. The information is scattered across different portals, is frequently presented in a limited set of languages, and requires users to know which service they need and how to navigate to it. The problem addressed by this project can be summarised as follows:

1. Citizens often struggle to find and understand government services, schemes, and application procedures.
2. Language barriers make it difficult for many people to access digital public-service information.
3. Existing government portals can be complex, especially for citizens with limited digital literacy.

4. Citizens need a simple way to ask questions about documents, eligibility, applications, complaints, and nearby service centres.
5. Therefore, a multilingual AI voice assistant is needed to understand citizens in their own language and provide simple, accessible, step-by-step guidance.

The proposed Multilingual Voice-Based Public Service Assistant lets a citizen simply speak a question, for example “How can I apply for a birth certificate?”, in Tamil, Hindi, English or another supported language. The system converts the speech to text, detects the language, understands the intent using AI and Natural Language Processing (NLP), identifies the relevant government service, retrieves verified service information, and responds with simple step-by-step guidance that is translated and spoken back in the citizen’s own language.

1.3 DIRECT BENEFITS

- **Multilingual support–** citizens can interact in their preferred Indian language.
- **Voice-based access –** there is no need for typing or complex navigation.
- **AI-powered understanding–** natural language questions and user intent are understood, not just keywords.
- **Easygovernmentserviceaccess –** the relevant public service is identified automa tic ally.
- **Step-by-stepguidance –** procedures, documents, fees and next steps are explained clear ly.
- **Voiceresponse –** answers are provided through speech in the user’s language.

1.4 INDIRECT BENEFITS

- **Improvedaccessibility–** useful for elderly people, low-literacy users and people who find text-based systems difficult.
- **Saves time –** reduces the effort needed to search through multiple portals or websites.
- **Simpleanduser-friendly –** designed for easy interaction with minimal technical knowledge.

- **Digital inclusion –** brings citizens who are uncomfortable with English or complex portals into the digital governance ecosystem.
- **Future integration –** can be connected with verified government APIs and databases for updated information.

1.5 ORGANIZATION OF THE REPORT

The remainder of this report is organised as follows. Chapter 2 reviews existing platforms and research related to multilingual and voice-based public-service delivery and identifies the research gap. Chapter 3 describes the proposed system, its workflow, modules, system requirements and technologies. Chapter 4 presents the results in terms of the stage-wise output of the system for a sample query and the extent to which the objectives are met. Chapter 5 concludes the report and outlines future trends. The appendix gives a sample interaction, and the references list the sources consulted.

CHAPTER 2

REVIEW OF LITERATURE

2.1 RELATED BACKGROUNDS

Several national platforms and research initiatives provide language technology or digital access to government services. The following systems were reviewed in this work.

1. BHASHINI (Digital India) provides speech recognition, text-to-speech and translation for multiple Indian languages. It mainly provides language technology; complete public-service assistance is not its sole focus.

2. UMANG provides multiple Central and State Government services through a single digital platform. Users may still need to identify the correct service and understand the procedure themselves.

3. AI4Bharat develops AI and speech technologies for Indian languages. Language processing alone does not provide complete government-service guidance.

4. Kumbh Sah'AI'yak is a multilingual, voice-enabled AI assistant for information and navigation during Maha Kumbh 2025. It is designed for a specific event rather than general government services.

5. The proposed system, a Citizen Voice Assistant, combines multilingual voice recognition, AI, government-service navigation and simple step-by-step guidance in one platform. Its future scope includes integration with verified real-time government APIs and databases.

Table 2.1 Summary of existing systems and their limitations

S.No	Existing System / Research	Technology / Contribution	Limitation
1	BHASHINI – Digital India	Speech recognition, text-to-speech and translation for multiple Indian languages	Mainly language technology; complete public-service assistance is not its sole focus
2	UMANG	Multiple Central and State Government services on a single digital platform	Users may still need to identify the correct service and understand the procedure themselves

S.No	Existing System / Research	Technology / Contribution	Limitation
3	AI4Bharat	AI and speech technologies for Indian languages	Language processing alone does not provide complete government-service guidance
4	Kumbh Sah'AI'yak	Multilingual, voice-enabled AI assistant for information and navigation during Maha Kumbh 2025	Designed for a specific event rather than general government services
5	Proposed System – Citizen Voice Assistant	Multilingual voice recognition + AI + government-service navigation + step-by-step guidance in one platform	Depends on verified service data; real-time API integration is a future scope

Existing systems such as BHASHINI, UMANG, AI4Bharat and Kumbh Sah'AI'yak provide multilingual language processing, digital services and voice-based assistance, but each addresses only a part of the overall problem.

2.2 EXISTING SYSTEM

The existing system refers to the currently available platforms, applications and technologies that provide citizens with government information or public services through digital, chatbot, voice or multilingual interfaces. These include:

- **Government service portals:** provide citizens access to various government schemes, certificates, applications and services online.
- **Mobile government apps:** allow users to access selected public services through smartphones.
- **AI chatbots:** answer frequently asked questions and provide basic service-related information.
- **Voiceassistants:** enable users to interact with digital services using spoken commands.
- **Multilingual translation systems:** help users access information in different Indian languages.

2.3 RESEARCH GAP

The review shows that language technology, service portals, chatbots and voice interfaces exist as separate components. Portals and apps expect the citizen to know which service to look for; language platforms do not by themselves guide a citizen through eligibility, documents and procedure; and event-specific assistants do not cover general public services. There is therefore a need for a single, voice-first platform that understands a citizen's question in their own language, identifies the correct service, and responds with simple step-by-step guidance in the same language. The proposed system is designed to fill this gap.

CHAPTER 3

SYSTEM IMPLEMENTATION

3.1 PROPOSED SYSTEM

The proposed system is a voice-first, multilingual assistant that guides citizens to public services. The new features that distinguish it from existing systems are:

- **Voice-first interaction** citizens can speak instead of typing.
- **Multiple Indian languages** supports communication in the user's preferred language.
- **AI-based query understanding** understands the citizen's intent, not just keywords.
- **Automatic service identification** finds the relevant government service from the query.
- **Step-by-step guidance** explains documents, procedure, fees and next steps simply.
- **Voice response in the samelanguage** gives the final guidance through natural speech.
- **Single integrated platform** combines voice, AI, language and public-service guidance in one system.

3.2 SYSTEM WORKFLOW

The assistant processes every request through the ten-stage pipeline shown in Figure 3.1.

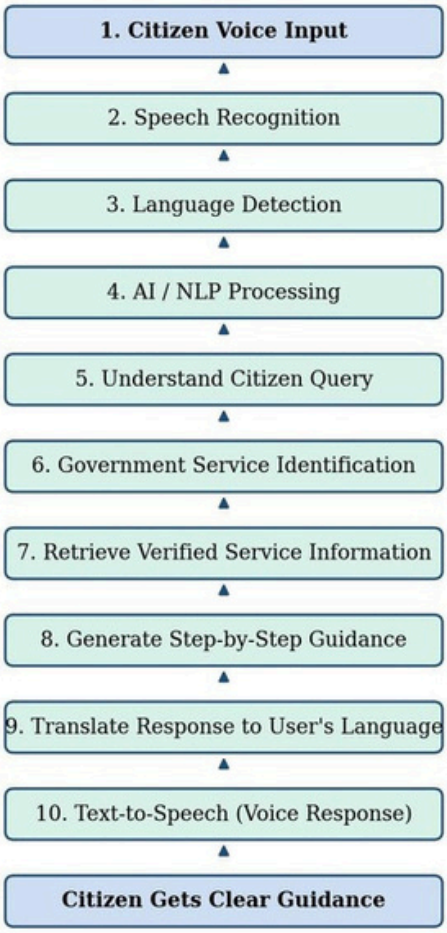


Figure 3.1 System workflow of the proposed multilingual voice-based public service assistant

1. Citizen voice input: the user speaks the query in their preferred language.
2. Speech recognition: the spoken words are converted into text accurately.
3. Language detection: the language of the user is identified and the context is set.
4. AI/NLP processing: the intent is understood, key information is extracted and the query is classified.
5. Government service identification: the relevant government service and department are found.
6. Retrieval of verified service information: accurate, updated details are fetched from trusted government databases or APIs.
7. Generation of step-by-step guidance: clear instructions, required documents, fees (if any) and timelines are produced in an easy-to-understand format.
8. Translation: the response is converted into the user's selected language.

9. Text-to-speech: the response is read aloud in the user's language with natural voice.
10. The citizen receives clear guidance – easy, accurate, in their own language, anytime and anywhere.

3.3MODULE DESCRIPTION

3.3.1 Voice Input and Speech Recognition Module

Captures the citizen's spoken query through a microphone and converts it into text using a speech recognition library or service. It forms the entry point of the system and enables users to interact without typing.

3.3.2 Language Detection Module

Identifies the language of the recognised text (for example Tamil, Hindi, English) so that subsequent processing and the final response can be handled in the same language. The language may also be selected manually.

3.3.3 AI/NLP Query Understanding Module

Uses Indian-language NLP models to interpret the meaning of the query, extract key information and classify the citizen's intent, for example "Birth Certificate application". This allows the assistant to understand questions phrased in different ways.

3.3.4 Service Identification and Information Retrieval Module

Maps the identified intent to the relevant government service and department and retrieves verified details such as eligibility, required documents, procedure, fees and timelines from the government service database, with provision for trusted APIs.

3.3.5 Guidance Generation Module

Converts the retrieved information into simple step-by-step guidance that a citizen with limited digital literacy can follow, covering documents, online and offline procedures and next steps.

3.3.6 Translation and Text-to-Speech Module

Translates the guidance into the user's language using a translation API or Indian-language translation models and reads it aloud through a text-to-speech service, so the citizen receives the answer as natural speech.

3.3.7 User Interface and Security Module

A web-based interface built with HTML, CSS and JavaScript lets users start voice interaction and view the text response. Authentication and secure HTTPS communication protect data transmission and access to verified government data.

3.4 SYSTEM REQUIREMENTS

The hardware,software, data/API and network requirementsare summarised in Table 3.1.

Table3.1Systemrequirementsoftheproposedsystem

Category	Requirement
Hardware	Processor – Intel i7 / Ryzen 5 or higher; RAM – 8 GB minimum; Storage – 256 GB SSD; Microphone and speaker; GPU – optional
Software	Windows / Linux; Python 3.8+; Flask / Django; HTML, CSS, JavaScript; speech recognition and NLP libraries; translation and text-to-speech APIs
Data and API	Government service database; Indian language models; translation API; text-to-speech API; authentication and security
Network	Stable internet connection; API connectivity; cloud hosting (optional); secure HTTPS connection

3.5 TECHNOLOGIES USED

- **Python(3.8orhigher)** as the programming language for the back end.
- **Flask/Django** as the web framework for the back end, with HTML, CSS and JavaScript (React / Angular optional) for the front end.
- **SpeechRecognition** library for converting speech to text.
- **NLP libraries** such as Hugging Face with Indian-language models such as IndicBERT for intent understanding.
- **Translation:** Google Translate API or IndicTrans for Indian-language translation.
- **Text-to-speech:** gTTS, Google TTS or Azure TTS for voice responses.
- **Cloudhosting (optional)** for scalability and reliability, with SSL/HTTPS for secure data transmission.

CHAPTER 4

RESULT

This chapter presents the output produced by theproposed system at each stage of the workflow, using the example query “How can I apply for a birth certificate?” spoken in Tamil, and then relates the system features to the objectives defined in Chapter 1.

4.1 SAMPLE QUERY WALKTHROUGH

Table 4.1 Stage-wise output for a sample citizen query

Stage	Output of the system
Voice input	Citizen speaks: “How can I apply for a birthcertificate?” (in Tamil)
Speech recognition	The spoken query is converted to text
Language detection	Language detected: Tamil
AI/NLP processing	Intent: Birth Certificate Service
Service identification	Service found: Birth Certificate (Department of Health & Family Welfare)
Retrieval of information	Required documents: ID proof, hospital record, application form.
Guidance generation	Procedure: online / offline (links, steps, timelines)
Translation	Simple step-by-step instructions listing documents, procedure, fees (if any) and timelines
Voice response	Response translated to Tamil (or the user’s language)
	Guidance spokenin the user’s languagewithnatural voice

The walkthrough shows that a single spoken question is sufficient for the citizen to obtain the correct service, the documents needed and the procedure, without searching through multiple portals or typing in English. The same pipeline applies to other categories such as schemes, healthcare, transport, education, jobs and complaints.

4.2 ACHIEVEMENT OF OBJECTIVES

Table 4.2 Mapping of project objectives to system features

S.No	Objective	How it is addressed

1	Multilingual AI voice assistant	Speech recognition, language detection, translation and text-to-speech in the user's preferred Indian language
S.No	**Objective**	**How it is addressed**
2	Understand voice-based queries on public services	AI/NLP intent understanding and automatic service identification
3	Simple, accurate step-by-step guidance	Retrieval of verified service information and guidance generation covering eligibility, documents, procedure and service centres
4	Reduce language and digitalliteracy barriers	Voice-first input and spoken responses; no typing or complex navigation needed
5	Accessible, user-friendly, anytime platform	Single integrated platform that can be hosted on the cloud for round-the-clock access

4.3 LIMITATIONS

- The accuracy of the guidance depends on the quality and currency of the government service data; integration with verified real-time government APIs and databases is a future scope.
- Speech recognition and translation quality can vary across languages, dialects and noisy environments.
- A stable internet connection is needed for API-based translation and text-to-speech services.
- Performance measures such as speech recognition accuracy, intent classification accuracy and response time are to be recordedduring testing with real users.

CHAPTER 5

CONCLUSION AND FUTURE TRENDS

5.1 CONCLUSION

Citizens often cannot benefit from government services because of language barriers, complex portals and limited digital literacy. This project presented a Multilingual VoiceBased Public Service Assistant that lets citizens ask questions by voice in their own

language and receive clear, step-by-step spoken guidance about eligibility, documents, procedures, fees and service centres. By integrating speech recognition, language detection, AI-based query understanding, service identification, translation and text-to-speech in a single platform, the system fills the gap left by existing portals, chatbots and language platforms that address only part of the problem.

The proposed assistant promotes accessibility, inclusiveness and efficiency in publicservice delivery, particularly for elderly and low-literacy users, and demonstrates how AI can be used to make digital governance truly citizen-centric.

5.2 FUTURE TRENDS

- Integration with verified real-time government APIs and databases so that eligibility, fees, application status and timelines are always current.
- Support for additional Indian languages and regional dialects, with improved recognition in noisy environments.
- Availability through mobile applications, messaging platforms and telephone (IVR) channels to reach users without smartphones or internet access.
- Personalised guidance based on the citizen's profile, including scheme eligibility checks and reminders.
- Support for complaint registration and application tracking through voice.
- Location-based guidance to nearby service centres and offices.
- Secure authentication and privacy-preserving handling of citizen data.

APPENDIX I

A. SAMPLE INTERACTION

The following shows a typical interaction with the assistant.

Step	Description
Citizen (voice, Tamil)	"How can I apply for a birth certificate?"
System – language detected	Tamil
System – intent	Birth Certificate Service (Department of Health & Family Welfare)

System – guidance	Required documents: ID proof, hospital record, application form. Procedure: online or offline, with links, steps and timelines.
System – voice response	The guidance above, translated into Tamil and spoken aloud

B. KEY FEATURES OF THE PROPOSED SYSTEM

- Voice-first, multilingual interaction.
- AI-based intent understanding and automatic service identification.
- Step-by-step guidance covering documents, procedure, fees and next steps.
- Voice response in the same language as the question.
- Single integrated platform with provision for verified government APIs.

REFERENCES

1. Ministry of Electronics & Information Technology (MeitY), Government of India – Digital governance and citizen-centric technology initiatives.
2. Digital India Programme – Digital transformation and delivery of public services.
3. BHASHINI – Multilingual AI, speech recognition, translation and text-to-speech.
4. UMANG – Integrated access to government services through a single platform.
5. AI4Bharat, IIT Madras – Indian-language NLP, speech and translation technologies.
6. National Informatics Centre (NIC) – Development of digital platforms and egovernance solutions.
7. Kumbh Sah'AI'yak – Multilingual AI-based information and voice assistance.